Chapter test

32 Regulation of body temperature

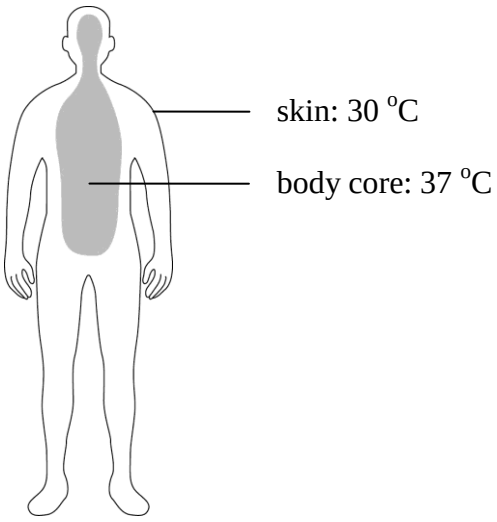
Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

Directions: Questions 1 and 2 refer to the diagram below, which shows the skin temperature and core body temperature of a person when the air temperature of the room he stayed in is maintained at 25 ^{o}C.



★ 1. Which of the following would be the possible skin temperature and core body temperature when the air temperature of the room dropped to 15 ^{o}C?

	skin temperature (^{o}C)	***core body temperature*** (^{o}C)
A.	25	37
B.	30	30
C.	30	20
D.	25	20

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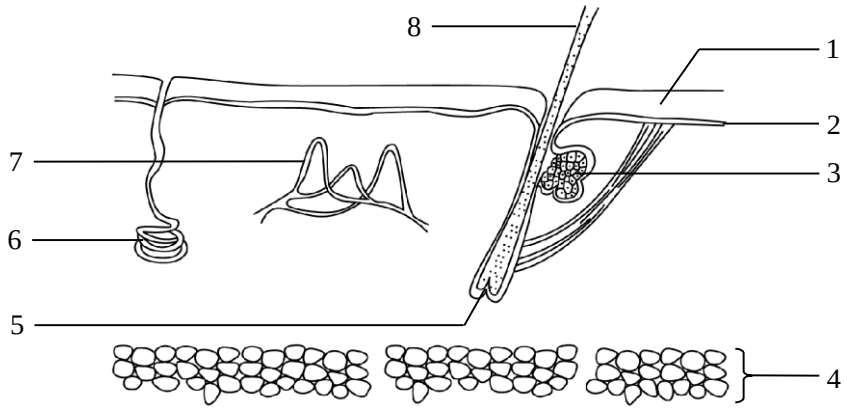
2. Which of the following is the possible response if the person stays at 15 ^{o}C for an hour without additional clothing?

A. The subcutaneous fat layer becomes thicker.
B. The growth rate of the hairs on the skin increases.
C. The blood vessels near the skin surface dilate.
D. The metabolic rate of skeletal muscles increases.

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Directions: Questions 3 to 5 refer to the diagram below, which shows a section of the human skin.



3. On a hot day, which of the following changes help regulate the body temperature?

(1) Structure 3 increases its secretion.
(2) Structure 7 dilates.
(3) Structure 8 lies flat.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

4. Which of the following structures help(s) defend against pathogens?

A. 1 only
B. 1 and 3 only
C. 1, 2 and 3 only
D. 4, 6 and 8 only ☐

5. Which of the following structures in the skin contain cells that actively undergo cell division?

A. 1 and 3
B. 2 and 5
C. 2 and 6
D. 5 and 7 ☐

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6. Which of the following are the changes in the skin when there is a sudden drop in air temperature?

	Erector muscles	***Arterioles in the skin***
A.	relax	dilate
B.	relax	constrict
C.	contract	dilate
D.	contract	constrict

☐

★ 7. Which of the following is/are regulated by negative feedback mechanism?

(1) putting on more clothes when feeling cold
(2) increased release of thyroxine under cold conditions
(3) lowered breathing rate under cold conditions

A. (2) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

☐

8. When a person moves from a sauna to an air-conditioned room, the following events occur.

1. Nerve impulses are sent to skeletal muscles.
2. The heat gain centre is stimulated.
3. Shivering occurs.
4. Temperature receptors are stimulated by the change in temperature.

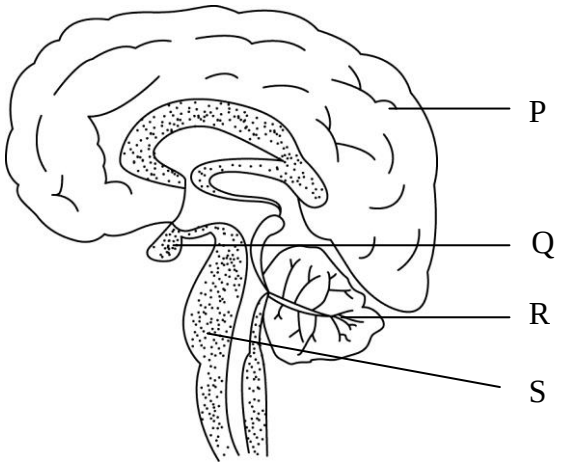
Which of the following shows the correct sequence?

A. 3 → 4 → 1 → 2
B. 2 → 1 → 4 → 3
C. 1 → 3 → 4 → 2
D. 4 → 2 → 1 → 3

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Directions: Questions 9 and 10 refer to the diagram below, which shows a section of the human brain.



★ 9. Which of the labelled part(s) is/are involved in body temperature regulation?

A. S only
B. P and Q only
C. Q, R and S only
D. P, Q, R and S ☐

★ 10. Structure Q of a person was damaged in a car accident. Which of the following could be a possible consequence?

A. loss of memory
B. loss of body balance
C. difficulty in breathing
D. unable to regulate body temperature ☐

Question	1	2	3	4	5	6	7	8	9	10
Related section	32.3	32.3	32.2	32.2	32.2	32.3	32.3	32.3	32.3	32.3

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B. Problem solving (20 marks)

11. A student compares the adaptive features to low temperature between Emperor Penguins and humans. Emperor Penguins live in the polar region where the temperature can be as low as -30 ^{o}C.

    (a) Adult Emperor Penguins are much larger than many other birds. Explain why a larger body size helps Emperor Penguins to survive at very low temperature. (2 marks)

    (b) Emperor Penguins are endotherms. They have soft feathers and a thick layer of subcutaneous fat to keep them warm. Suggest how these adaptive features help the penguins to maintain their body temperature. (2 marks)

    (Total: 4 marks)

12. Peter went jogging on a hot summer day.

    (a) After jogging for 30 minutes, his face became red and he sweated heavily.

        (i) Explain why his face became red. (4 marks)

        (ii) State the significance of these responses. (1 mark)

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(b) Peter forgot to bring water. After 90 minutes of jogging, his sweat production significantly reduced and his body temperature increased. He started to show heat stroke symptoms, such as feeling dizzy and having difficulty in breathing normally.

(i) Account for the increase in his body temperature. (2 marks)

(ii) Suggest ***one*** way he should do immediately to relief the heat stroke symptoms. (1 mark)

(c) When Peter went back home, he drank a bottle of sport drink. Explain why drinking sport drink is better than drinking water after heavy sweating. (2 marks)

(Total: 10 marks)

★ 13. An investigation was carried out to study the effect of drinking iced water on heat loss from the body. A man sat in a hot and dry room for 20 minutes. He was then asked to drink some iced water and remain seated for another 20 minutes. Temperature sensors were used to measure the temperature in his ear and skin. The table below shows the results.

Time (minute)	Temperature in the ear (^{o}C)	Skin temperature (^{o}C)
0	37.5	36.4
5	37.5	36.5
10	37.6	36.5
15	37.6	36.5
20	37.6	36.5
21	**Drinking iced water**	
25	37.4	37.0
30	37.0	37.6
35	37.3	37.2
40	37.6	37.0

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(a) Suggest why the temperature in the ear can roughly represent the core temperature. (1 mark)

(b) Describe the relationship between the skin temperature and core temperature of the man after drinking iced water. (2 marks)

(c) The blood flowing through the stomach reaches the brain within a minute. Explain how the change in skin temperature was brought about after drinking iced water. (3 marks)

(Total: 6 marks)

C. Exam challenge (10 marks)

★★ 14. The table below shows various means of heat loss from the skin at three different air temperatures and their relative proportions of the total heat loss.

Air temperature (^{o}C)	Relative proportion of the total heat loss (%)		
	Conduction and convection	Radiation	Evaporation
22	6	76	18
28	26	49	25
34	8	5	87

(a) Suggest how heat loss by conduction from the skin occurs. (1 mark)

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(b) What is the major means of heat loss from the skin at 22 ^{o}C? Describe how body heat generated in the liver is lost by this means. (4 marks)

(c) (i) Describe the physiological process occurring in the blood vessels in the skin when the air temperature is at 22 ^{o}C. Explain why this response is effective in reducing heat loss at 22 ^{o}C. (2 marks)

(ii) Describe the physiological process occurring in the blood vessels in the skin when the air temperature is at 34 ^{o}C. Explain why this response is ***not*** effective in promoting heat loss at 34 ^{o}C. (2 marks)

(d) How is the increase in heat loss by evaporation at 34 ^{o}C brought about? (1 mark)

(Total: 10 marks)

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Marking scheme

A. Concept checking (10 marks, 1 mark each)

1.	A	2.	D	3.	C	4.	B	5.	B
6.	D	7.	A	8.	D	9.	B	10.	D

(1 mark each)

B. Problem solving (20 marks)

11. (a) A larger body size reduces the surface area to volume ratio. (1)
Hence the rate of heat loss to the surroundings is reduced. (1)
(b) Dense feathers can trap air which is a good insulator of heat. (1)
Subcutaneous fat is also a good insulator of heat. (1)
Both features help reduce heat loss to the surroundings.

(Total: 4 marks)

12. (a) (i) During jogging, increased muscular contractions generated a large amount of heat. /
Peter absorbed heat from the sun. (1)
His body temperature increased and this stimulated the heat loss centre. (1)
The heat loss centre caused the arterioles in the skin to dilate. (1)
The blood flow to the skin surface increased. (1)
Therefore his face became red.
(ii) To promote heat loss to the surroundings in order to restore the increased body temperature (1)
(b) (i) The heat loss through evaporation of sweat decreased. (1)
The rate of heat gain is greater than the rate of heat loss from his body. (1)
Therefore, there was a net heat gain.
(ii) Rest in shaded area. /
Loosen his clothing. /
Drink water. (1)
(c) Sport drink contains mineral ions. (1)
It can replenish the minerals lost in sweat. (1)

(Total: 10 marks)

13. (a) The position of measurement is near to the brain. (1)
(b) As the core temperature decreases between 25 and 30 minutes, the skin temperature increases. (1)
As the core temperature increases between 30 and 40 minutes, the skin temperature

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decreases. (1)

(c) Iced water lowered the blood temperature at the stomach. (1)
The blood from the stomach reached the hypothalamus which detected the fall in blood temperature. (1)
Heat gain centre sent fewer nerve impulses to the sweat glands in the skin to produce less sweat / the arterioles in the skin to constrict / the erector muscles to contract. (1)
The rate of heat loss from the skin decreased.

(Total: 6 marks)

C. Exam challenge (10 marks)

14. (a) It occurs when the skin touches an object of lower temperature. (1)

(b) Radiation (1)
Body heat generated in the liver is transported to the skin by the blood. (1)
When blood flows through the capillaries near the skin surface, (1)
heat is lost from the blood to the surroundings by infrared radiation. (1)

(c) (i) Vasoconstriction / Arterioles constrict (1)
At 22 ^{o}C, the temperature gradient between the skin and the surroundings is steep.
The rate of heat loss by radiation would be high if the blood flow to the skin surface is not reduced. (1)

(ii) Vasodilation / Arterioles dilate (1)
At 34 ^{o}C, the temperature gradient between the skin and the surroundings is not steep.
The rate of heat loss by radiation is low even when the blood flow to the skin surface is increased. (1)

(d) Sweat glands secrete more sweat. (1)

(Total: 10 marks)

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